

Chapter 10 (Part 2) - Capacitor Charge & Discharge Using Square Signal LFG

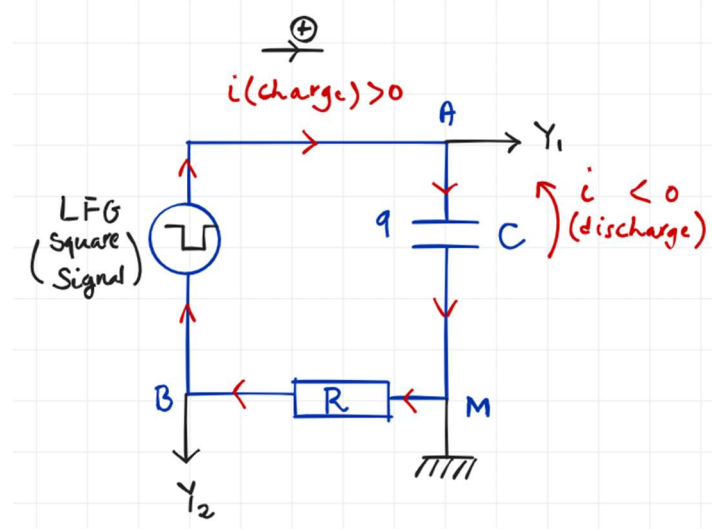
On channel 1 the oscilloscope allows to visualize the voltage $u_{y_1-Mass} = u_{AM} = u_c$

On channel 2 the oscilloscope allows to visualize the voltage $u_{y_2-Mass} = u_{BM} = -u_R = -R \cdot i$

To visualize the voltage $u_{MB} = u_R$, the reverse button must be operated on channel 2.

The 2 voltages visualized on the oscilloscope screen are then:

$u_{AM} = u_c$ and $u_{MB} = u_R$ (they have same direction)



Remark: When charging and discharging a capacitor using an LFG, we choose the direction of the current during the charge as the positive direction so:

$i \text{ (charge)} > 0$ and $i \text{ (discharge)} < 0$. As result, in all cases: $i = \frac{dq}{dt}$

During the charge q increases with t , so $\frac{dq}{dt} > 0$;

During the discharge q decreases with t , so $\frac{dq}{dt} < 0$.

The generator delivers a square voltage, where

$$u_G = \begin{cases} E & \text{for } 0 \leq t \leq \frac{T}{2} \\ 0 & \text{for } \frac{T}{2} \leq t \leq T \end{cases}$$

1. Capacitor charge:

$$u_G = E \text{ (for } 0 \leq t \leq \frac{T}{2})$$

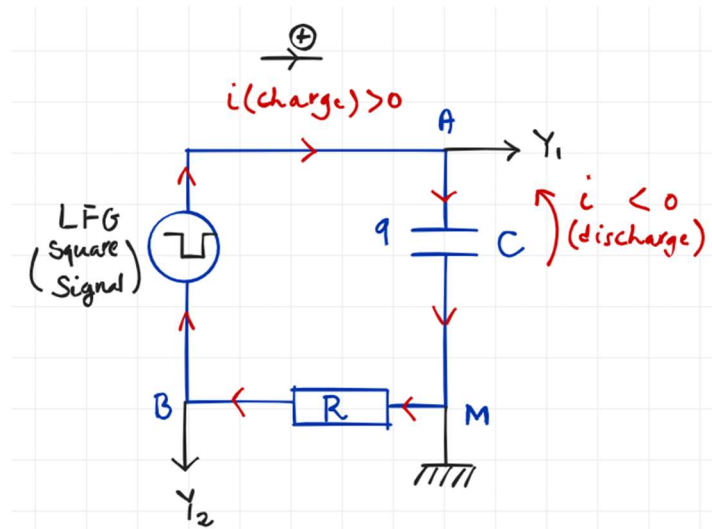
By the law of addition of voltages:

$$u_{AB} = u_{AM} + u_{MB} \text{ so } u_G = u_c + u_R \text{ so } E = u_c + R \cdot i$$

$$\text{Or } i = \frac{dq}{dt} = \frac{d}{dt}(Cu_c) = C \cdot \frac{du_c}{dt} \Rightarrow E = u_c + R \cdot C \frac{du_c}{dt}$$

The solution of this differential equation is:

$$u_c = E(1 - e^{-\frac{t}{\tau}}) \text{ where } \tau = R \cdot C$$



2. Capacitor discharge:

$$u_G = 0 \quad (\text{for } \frac{T}{2} \leq t \leq T)$$

By the law of addition of voltages:

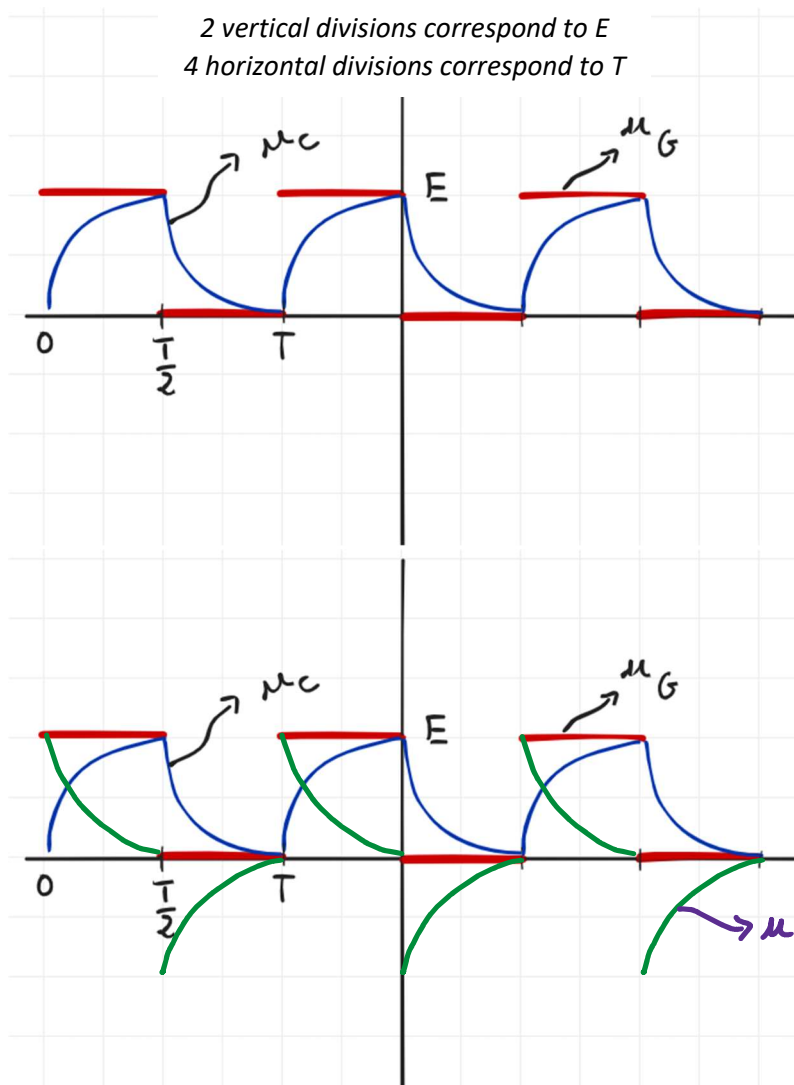
$$u_{AB} = u_{AM} + u_{MB} \text{ so } u_G = u_c + u_R \text{ so } 0 = u_c + R \cdot i$$

$$\text{Or } i = \frac{dq}{dt} = \frac{d}{dt}(Cu_c) = C \cdot \frac{du_c}{dt} \Rightarrow 0 = u_c + R \cdot C \frac{du_c}{dt}$$

The solution of this differential equation is:

$$u_c = E e^{-\frac{t}{\tau}} \text{ with } \tau = R \cdot C$$

Shape of the oscillograms obtained on the screen of the oscilloscope



The condition to visualize the phenomenon of complete charge:

$$5\tau \leq \frac{T}{2}$$

$$\Rightarrow T \geq 10\tau$$

$$\text{or } T \geq 10RC$$

Capacitor Charge Using Square Signal LFG ($0 \leq t \leq \frac{T}{2}$; $u_G = E$)	Capacitor Discharge ($\frac{T}{2} \leq t \leq T$; $u_G = 0$)
<i>Differential Equation in u_C</i>	
$\frac{du_C}{dt} + \frac{u_C}{RC} = \frac{E}{RC}$	$\frac{du_C}{dt} + \frac{u_C}{RC} = 0$
<i>Differential Equation in q</i>	
$\frac{dq}{dt} + \frac{q}{RC} = \frac{E}{R}$	$\frac{dq}{dt} + \frac{q}{RC} = 0$
<i>Differential Equation in i</i>	
$\frac{di}{dt} + \frac{i}{RC} = 0$	$\frac{di}{dt} + \frac{i}{RC} = 0$
<i>Differential Equation in u_R</i>	
$\frac{du_R}{dt} + \frac{u_R}{RC} = 0$	$\frac{du_R}{dt} + \frac{u_R}{RC} = 0$
<i>Expression of u_C in term of time</i>	
$u_C(t) = E(1 - e^{-\frac{t}{RC}})$	$u_C(t) = E e^{-\frac{t}{RC}}$
<i>Expression of q in term of time</i>	
$q(t) = Cu_C(t) = CE(1 - e^{-\frac{t}{RC}})$	$q(t) = Cu_C(t) = CE e^{-\frac{t}{RC}}$
<i>Expression of i in term of time</i>	
$i = \frac{dq}{dt} = C \frac{du_C}{dt} = \frac{E}{R} e^{-\frac{t}{RC}}$	$i = \frac{dq}{dt} = C \frac{du_C}{dt} = -\frac{E}{R} e^{-\frac{t}{RC}} < 0$
<i>Expression of u_R in term of time</i>	
$u_R = R.i = R.\frac{E}{R} e^{-\frac{t}{RC}} = E e^{-\frac{t}{RC}} > 0$	$u_R = R.i = R\left(-\frac{E}{R} e^{-\frac{t}{RC}}\right) = -E e^{-\frac{t}{RC}} < 0$